

Republic of the Philippines
CAVITE STATE UNIVERSITY
SILANG CAMPUS



Emerging Technologies Report

DCIT 26 – Application Development and Emerging Technologies

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Introduction

Emerging technologies describe newly developed technologies as well as the creative application of existing ones.

Characteristics of Emerging Technologies

Characteristics	Description
Radical novelty	Using existing technology innovatively.
Relatively fast growth	Characterized by learning, funding, and production
Coherence	Convergence of research streams forming a unified identity.
Prominent impact	Transforms businesses, industries and society.
Uncertainty and ambiguity	The outcomes of emerging technologies remain unpredictable.

Emerging Technology Areas

Augmented Reality (AR) - Real-world views with computer-generated overlays. Application areas are in-car navigation systems, shops, and computer games.
Artificial Intelligence (AI) – Simulation of human intellectual processes such as learning and reasoning.
Machine Learning – A type of AI that allows computers to learn from data. It is used in drug discovery, finance, and transport.
Quantum Computing - Uses qubits to solve problems. It is mainly developed by IBM and Google.

The Fourth Industrial Revolution

Stages of Industrial Revolution

Industry 1.0 – Introduction of mechanical production using steam and water power.
Industry 2.0 - Electricity and mass production.
Industry 3.0 – The era of automation using computers and electronics
Industry 4.0 - Era of IoT, robotics, VR/AR, and AI.

Building Blocks of Industry 4.0

- **Big Data Analytics -** Real-time analysis of data.
- **Internet of Things (IoT) –** Connecting objects to the internet for communication.
- **Cloud Computing –** Creating and using hosted services through the internet to make data-driven decisions.
- **Augmented Reality (AR) -** Supporting workers in warehouses and logistics.

Conclusion

Emerging technologies have distinct characteristics, which include novelty, rapid growth, coherence, impact and uncertainty. Moreover, they are integral components of Industry 4.0 and important in transforming businesses and societies.

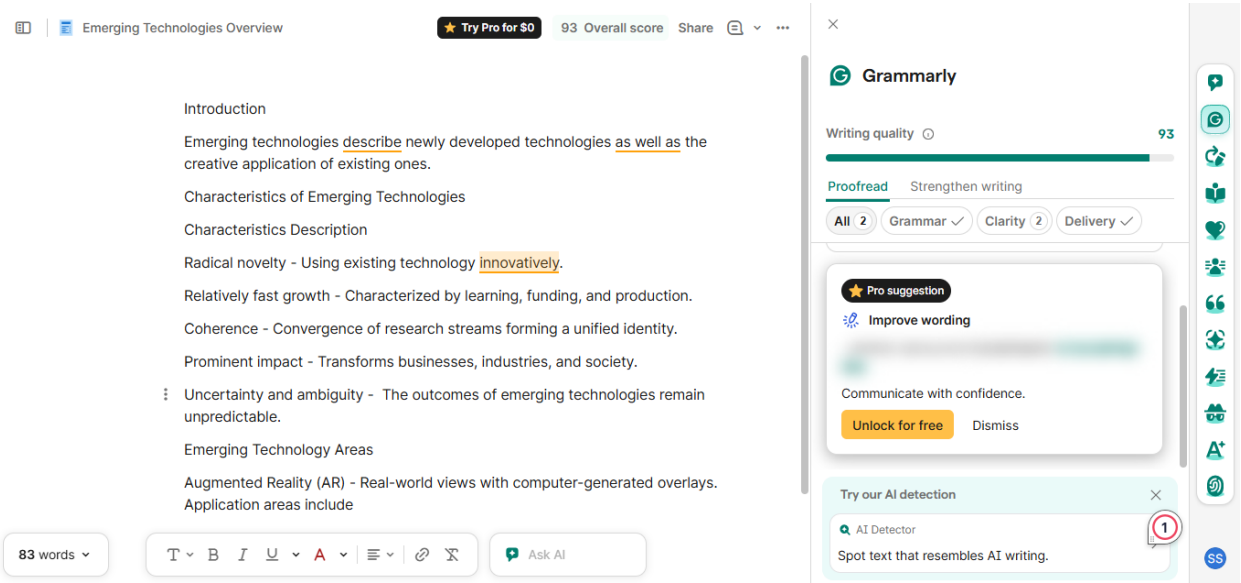
References

Module 1 Lecture Notes – DCIT 26 Application Development and Emerging Technologies

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BAGONG PILIPINAS

Grammarly Result



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